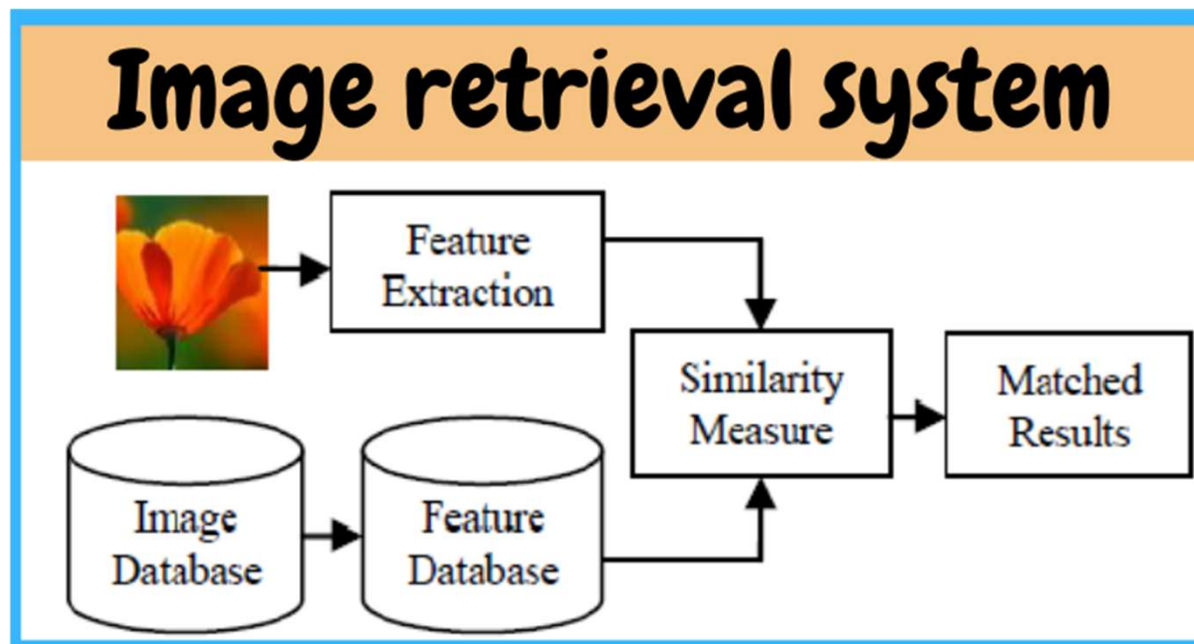


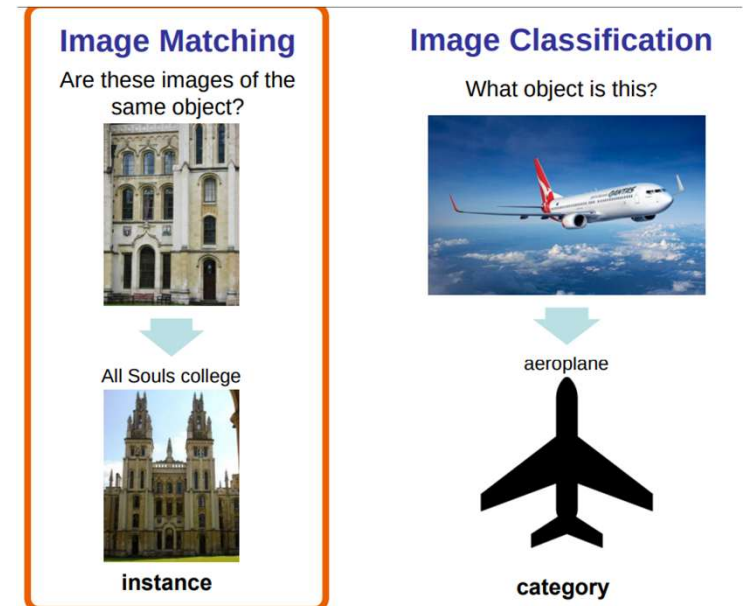
Matching in Embedding Spaces Using Deep Networks

Luke Jansen



Matching vs Classification

- Matching: Given an image, compute embedding and match it to the most similar image in a dataset
- Classification: Given an image, pick the most *likely* class based on learned weights

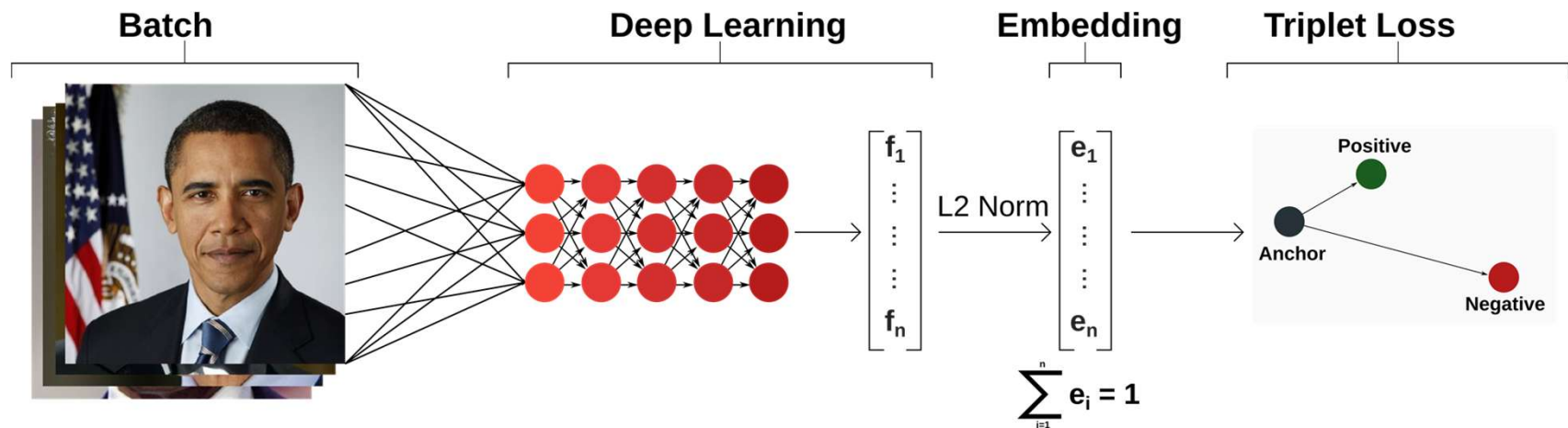


https://www.robots.ox.ac.uk/~vgg/projects/see_bibyte/events/overview_az.pdf

Case Study

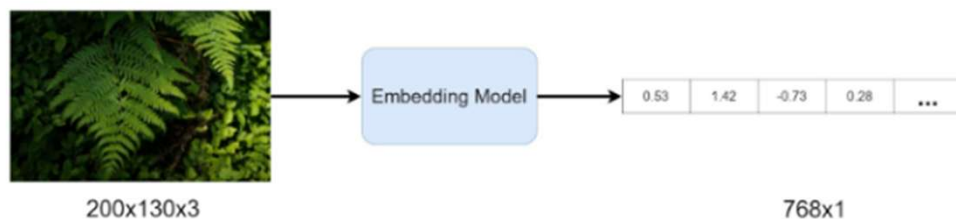
•FaceNet – Google, 2015

- Recognize input face against database of millions of faces
- Classification impossible
- 128-dimensional CNN embedding, triplet loss, semi-hard negative mining
- Validates match if distance is lower than threshold

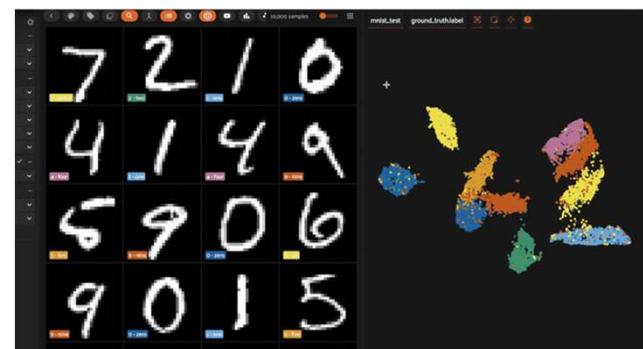


What is an Embedding?

- Simplify high dimensional image data into a vector that captures only the essential features of an image.
- Projects images into same space, where matching can be done
- Classical and deep network approaches



<https://dagshub.com/blog/image-embedding-benefits-use-cases-and-best-practices/>



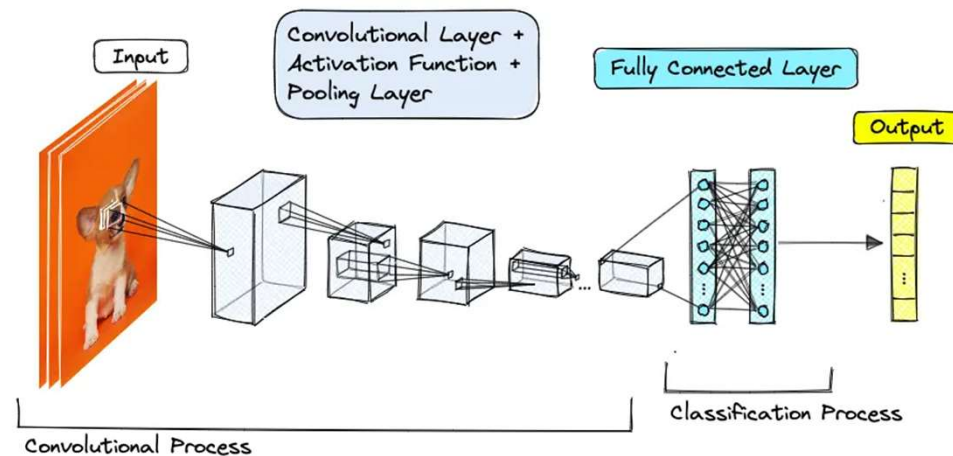
https://docs.voxel51.com/tutorials/image_embeddings.html

Crafting Feature/Embedding Space

- Classical: Hand pick filters, activations, run image through, output is embedding. Single pass.
- Classical ML: Hand pick filters, *learns* output weights.
- Deep Network: (usually) hand pick layer architecture, *learns* filter weights *and* output weights through back prop., output is embedding space where same-label images appear close together

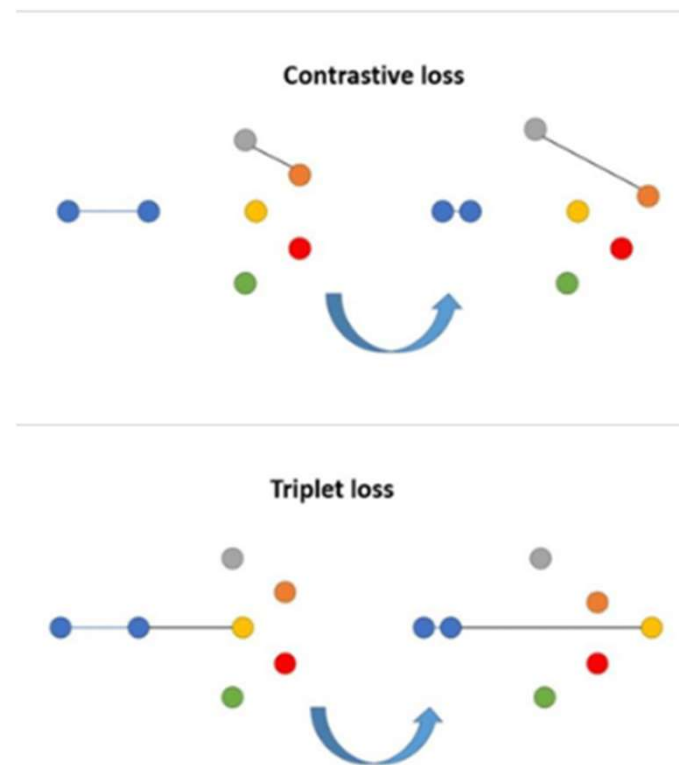
CNNs

- Convolution layers learn spatial features like edges, textures, etc.
- Pooling layers condense image so that increasingly higher level features can be learned



Deep Network Loss Functions

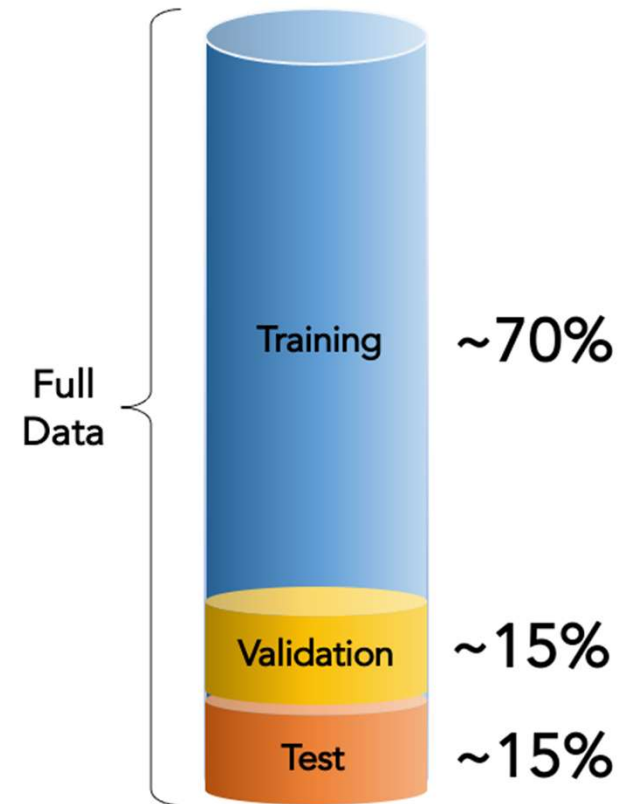
- Contrastive Loss: For each image pair, pull/push away based on match
- Triplet loss: More forgiving than contrastive loss. Catch: most random triplets already satisfy this, so hard negative mining is needed



https://www.researchgate.net/figure/sual-representation-of-a-gradient-descent-step-for-each-of-the-compared-losses_fig1_333418402

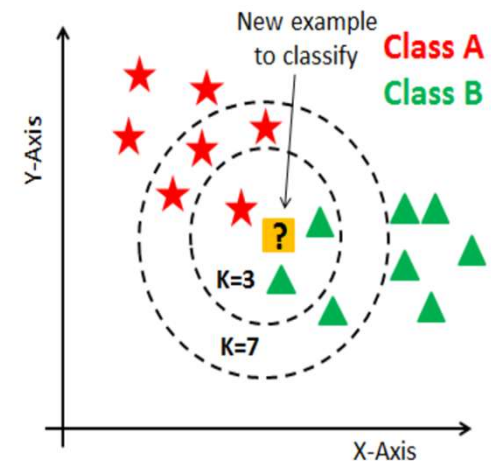
The Database & Gallery

- This database concept is the same for classification problems.
- Loss function changes role of the train dataset
- Embedding networks have separate “Gallery database



Matching

- Instead of predicting a label based on a softmax over the output layer (classification), predict label by finding best match with dataset
- Nearest neighbor/K-nearest neighbor
 - Find the closest neighbor(s)
- Nearest centroid



Summary

- Deep networks match by turning the problem into a geometrical one
- Network first learns how to embed image with train set
- Nearest neighbor search over gallery embeddings